

2.6 Natural Polymers

by Zeynep Guven

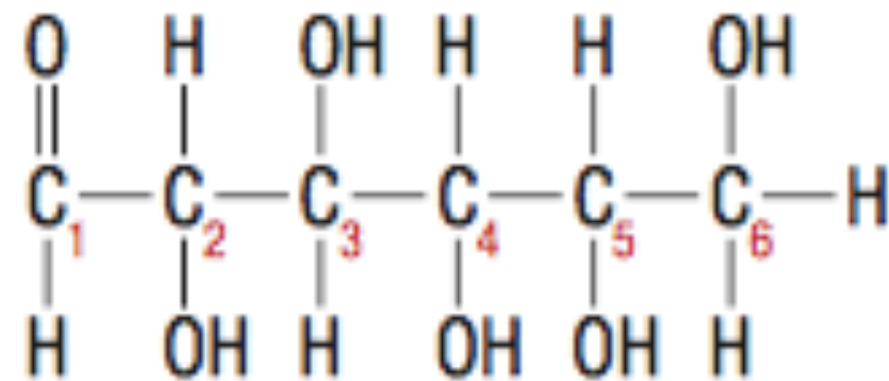
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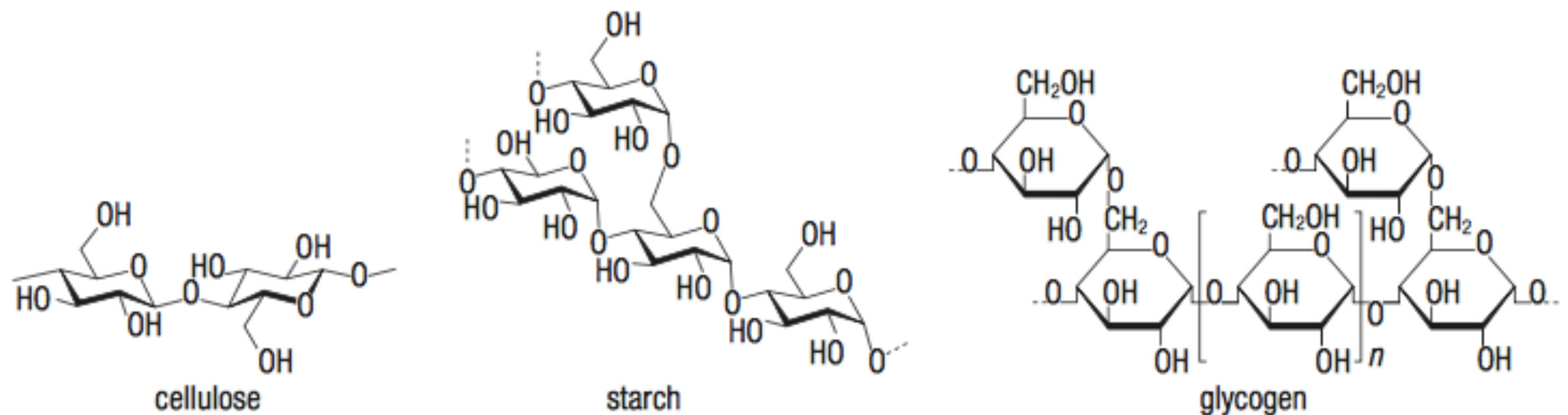
Nile Academy

November, 1st 2017

Polysaccharides

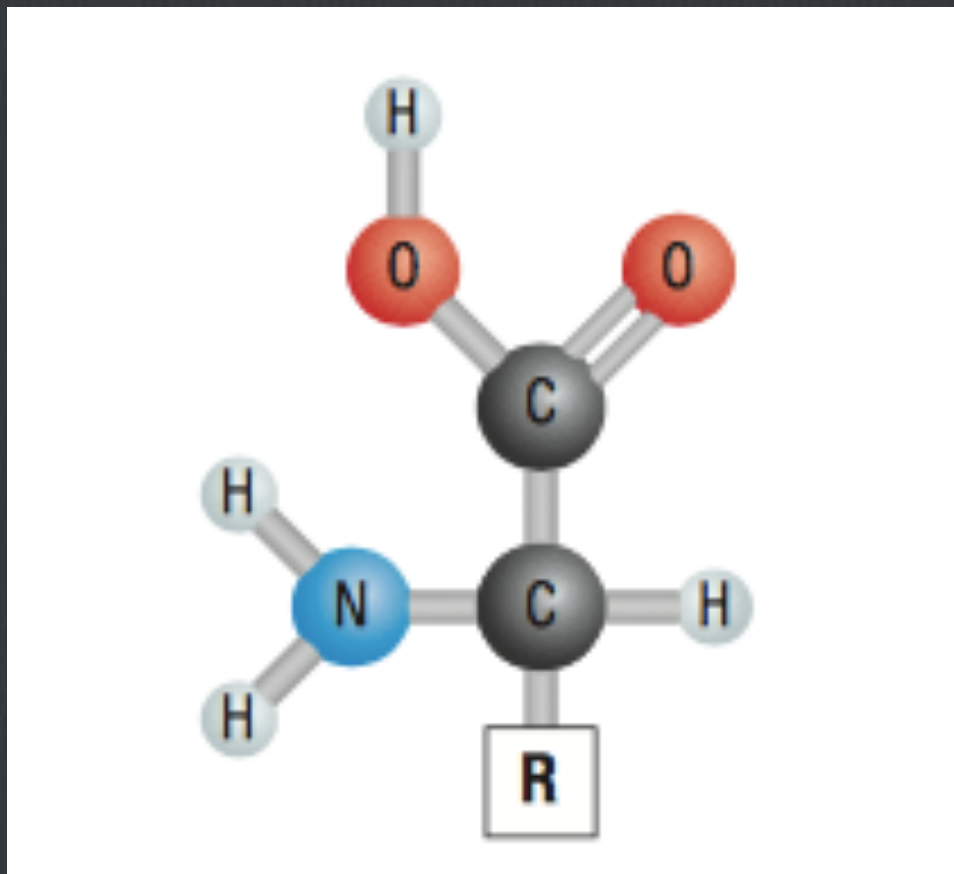


A **carbohydrate**, such as starch or cellulose, is a polymer of monosaccharides, or simple sugars. A **monosaccharide** is a ketone or aldehyde with many additional hydroxyl substituents. For example, glucose has a hexagonal structure with five hydroxyl groups.



When monosaccharide rings join together, they form large polymers consisting of many monosaccharide units. Such a polymer is called a **polysaccharide**. The third important polysaccharide, along with cellulose and glycogen, is starch.

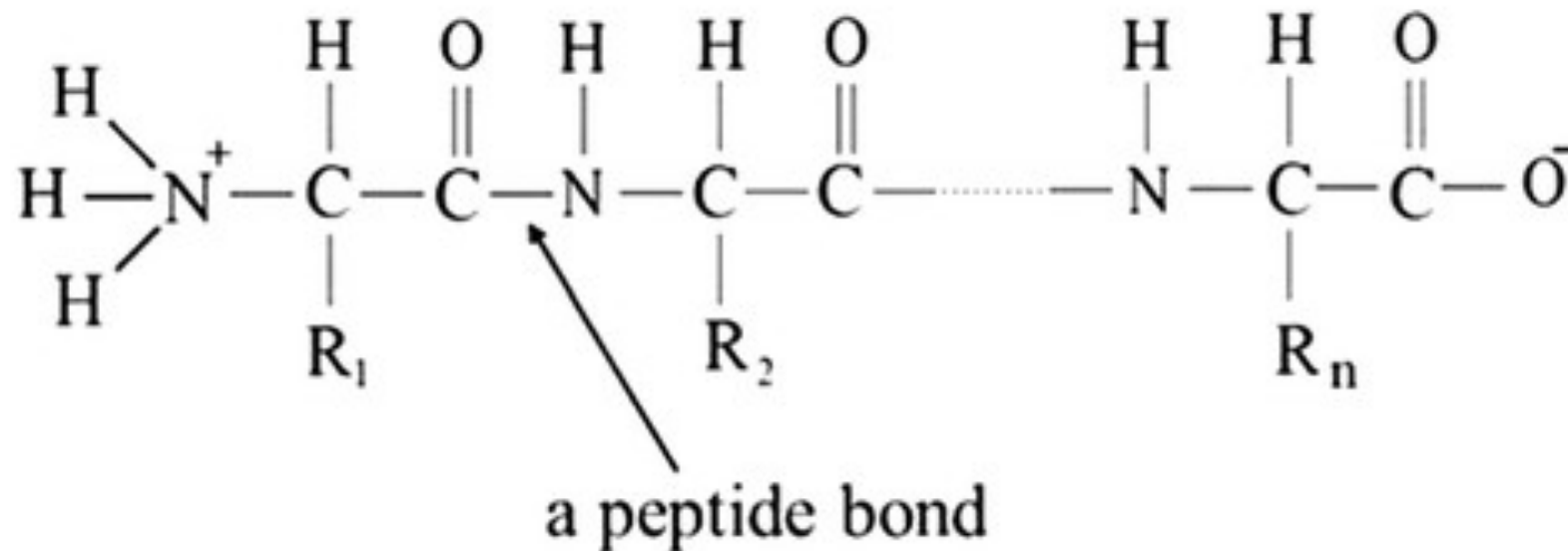
Proteins



All proteins are made of amino acids. An **amino acid** is an organic molecule that contains a carboxyl group ($-\text{COOH}$), an amino group ($-\text{NH}_2$), and a hydrogen atom all attached to the same carbon atom.

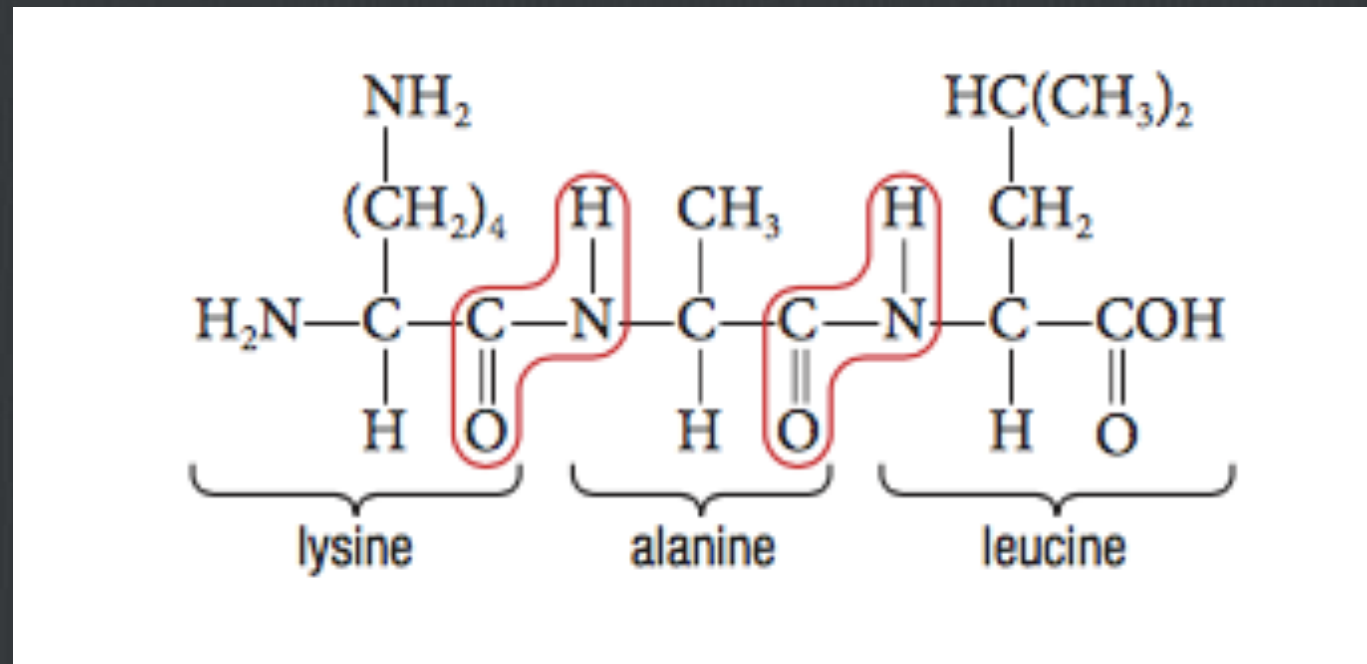
The fourth bond on that carbon atom links to an additional group or chain of atoms, generically designated as **R**. The nature of this group **gives each amino acid its distinct properties.**

Peptide



A protein polymer is built by condensation reactions between amino acids. A **peptide** is two or more amino acids linked together through a **condensation reaction between the amine and acid groups**.

Protein Structure

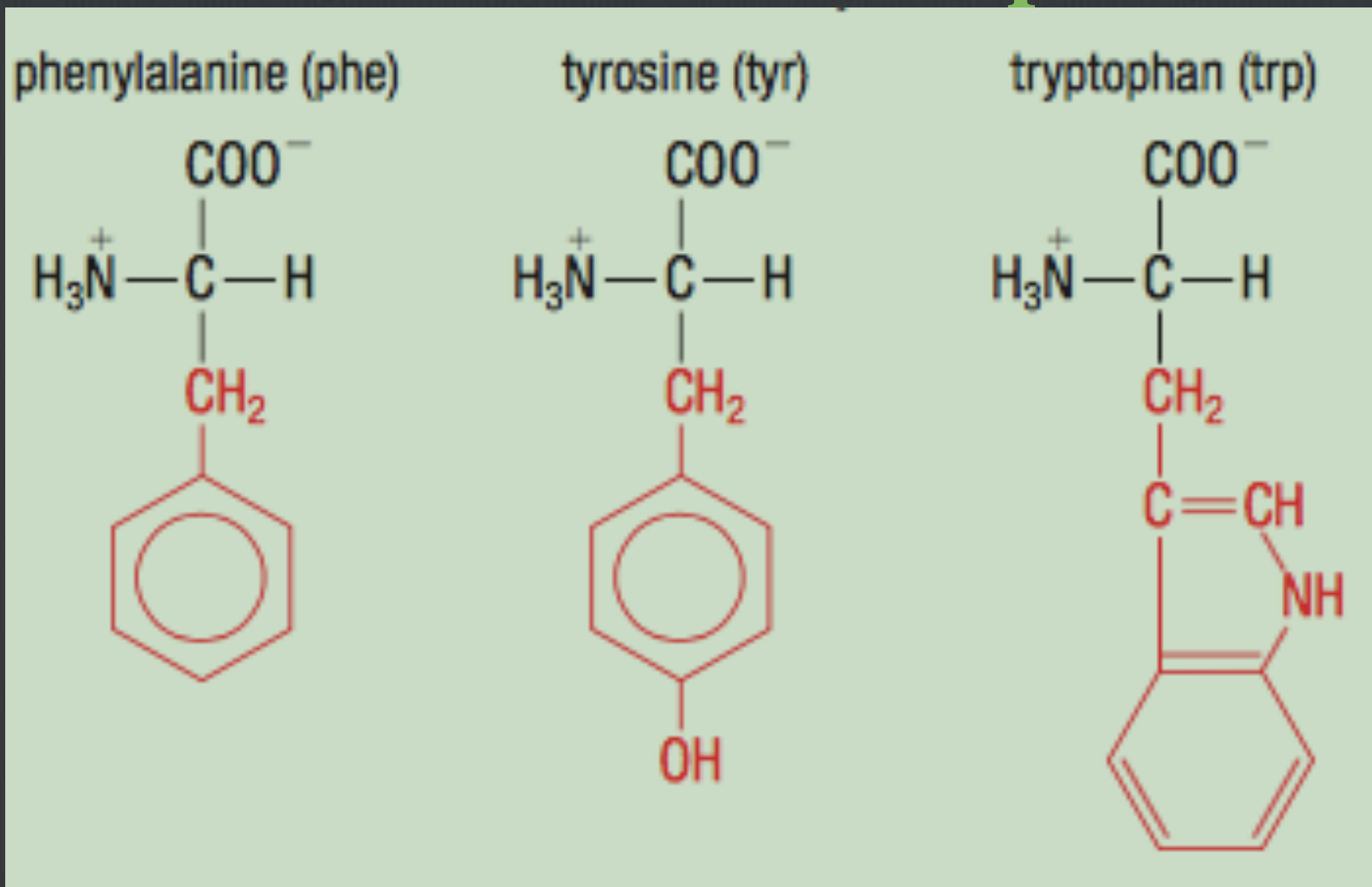


The sequence of amino acids in the protein chain is called the primary structure. Bio-chemists use the convention that the terminal carboxyl group is on the right and the terminal amino group is on the left.

For example, one possible sequence for a **tripeptide** (a chain of three amino acids joined by peptide bonds) containing lysine, alanine, and leucine.

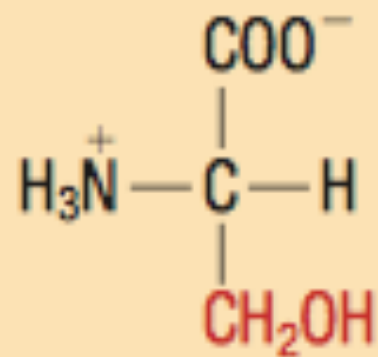
The 20 amino acids all have different side chains (R groups), **shown in red**.

Aromatic R Groups

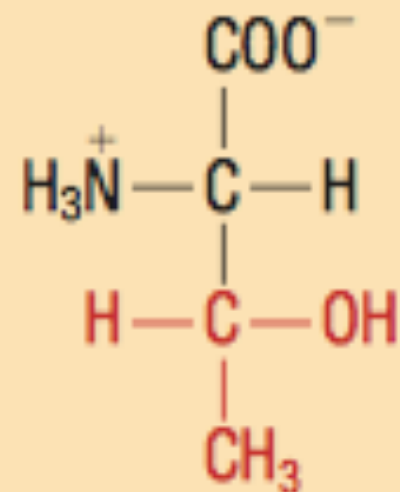


Polar, Uncharged R Groups

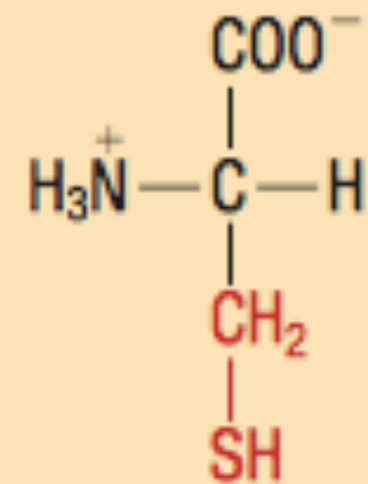
serine (ser)



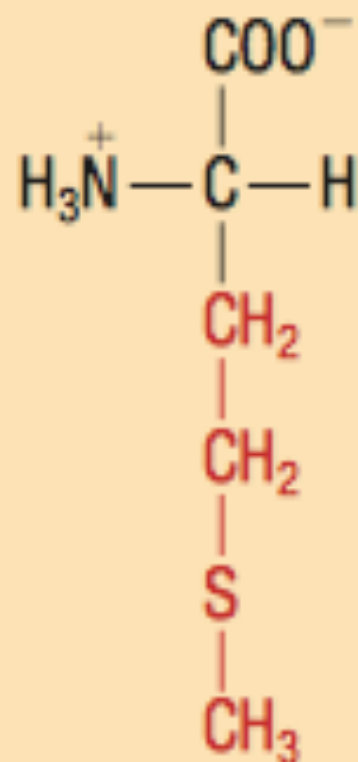
threonine (thr)



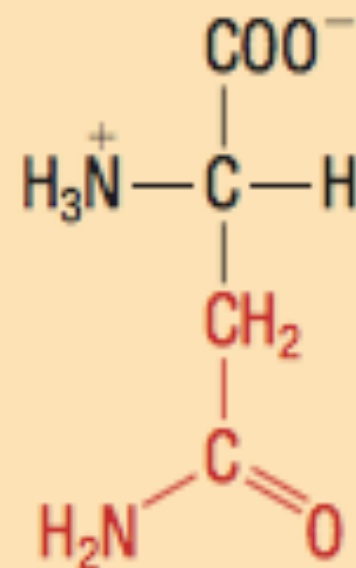
cysteine (cys)



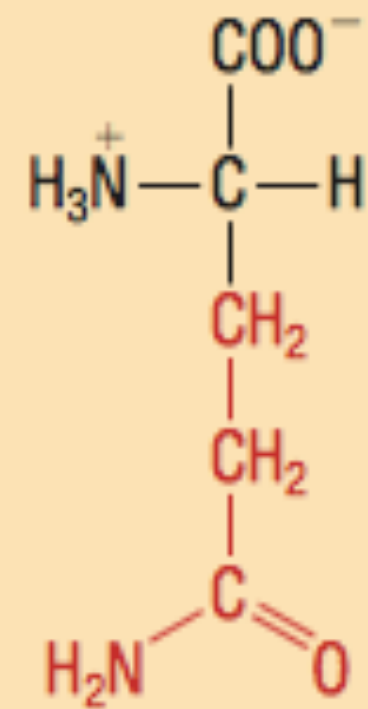
methionine (met)



asparagine (asn)

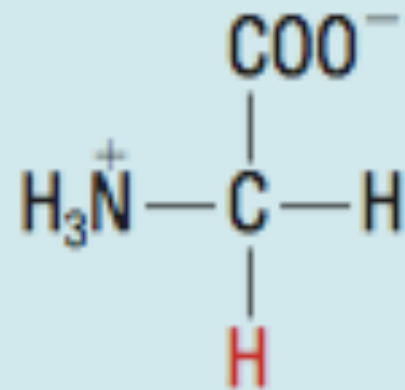


glutamine (gln)

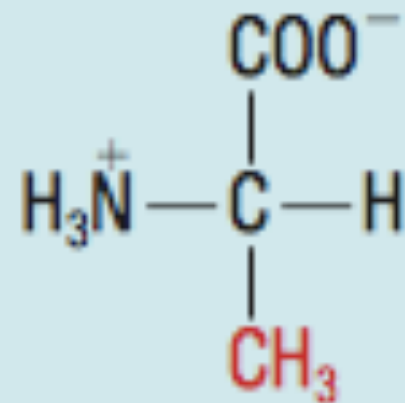


Non-polar, Aliphatic R Groups

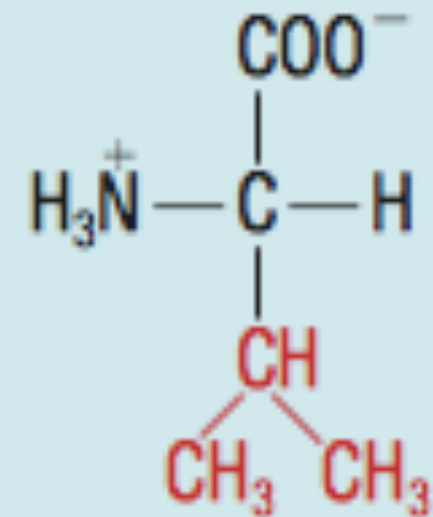
glycine (gly)



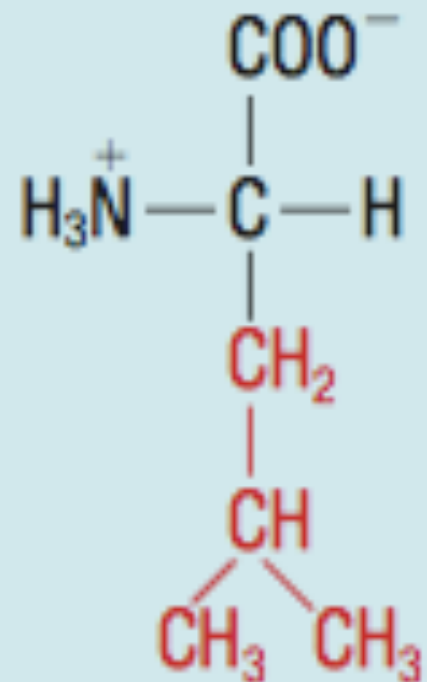
alanine (ala)



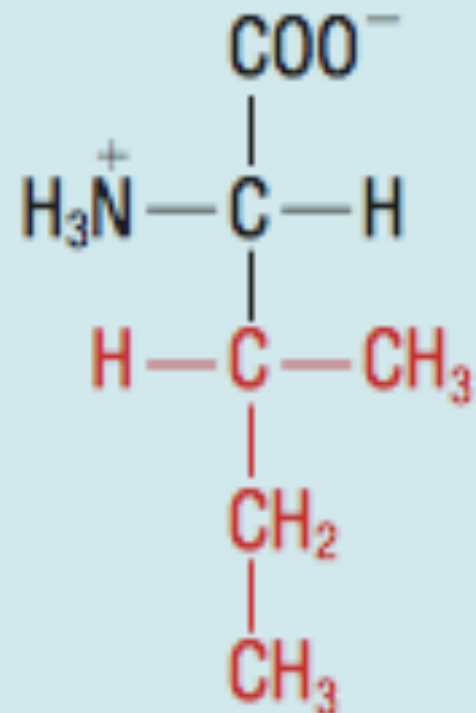
valine (val)



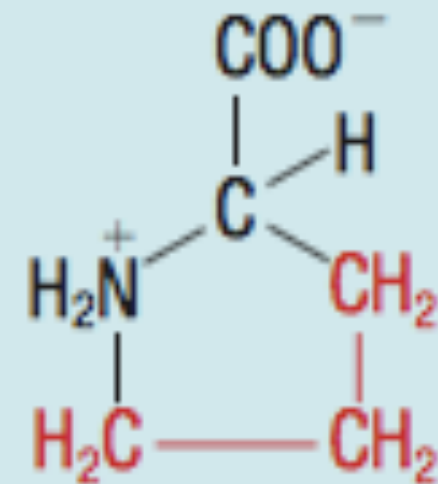
leucine (leu)



isoleucine (ile)

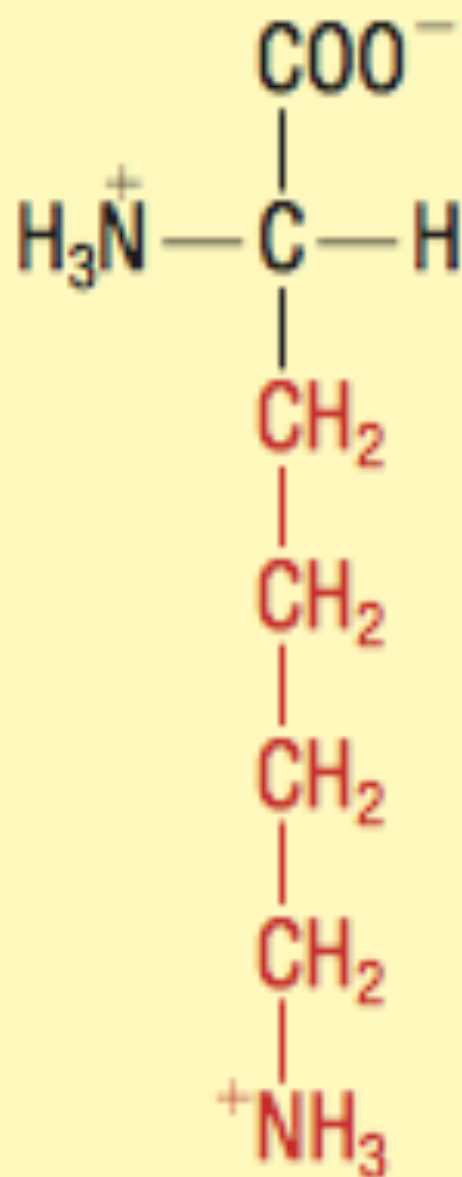


proline (pro)

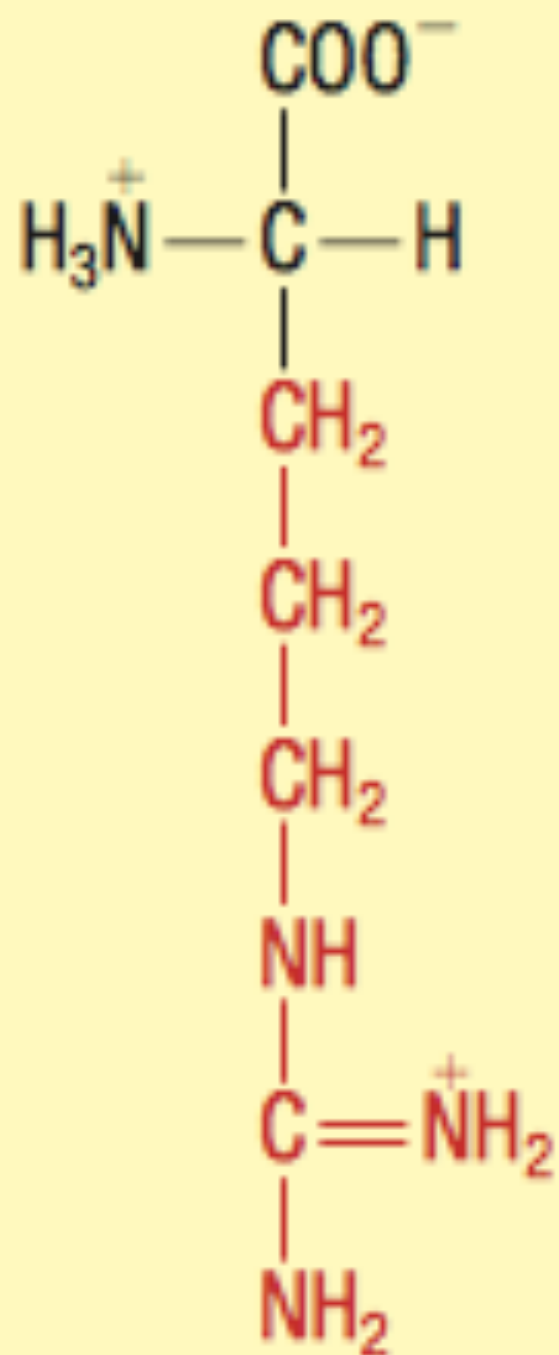


Positively Charged R Groups

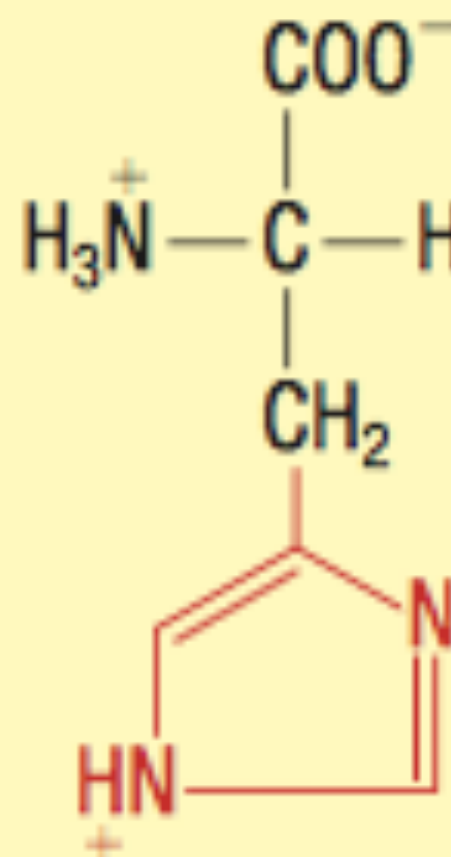
lysine (lys)



arginine (arg)

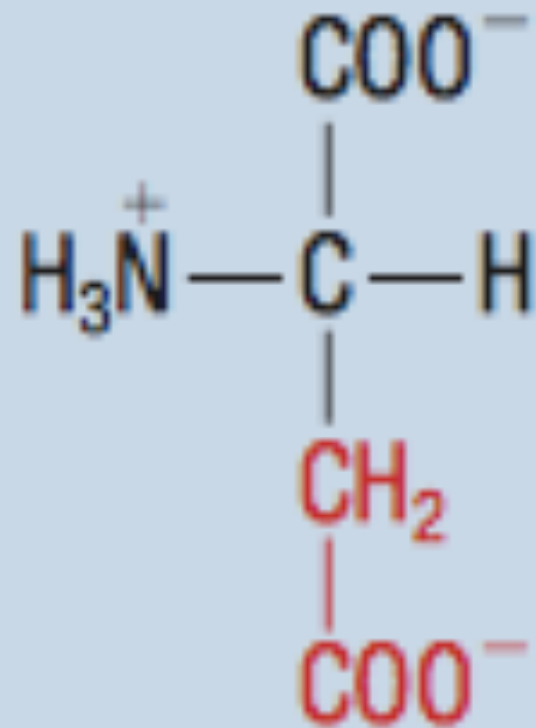


histidine (his)

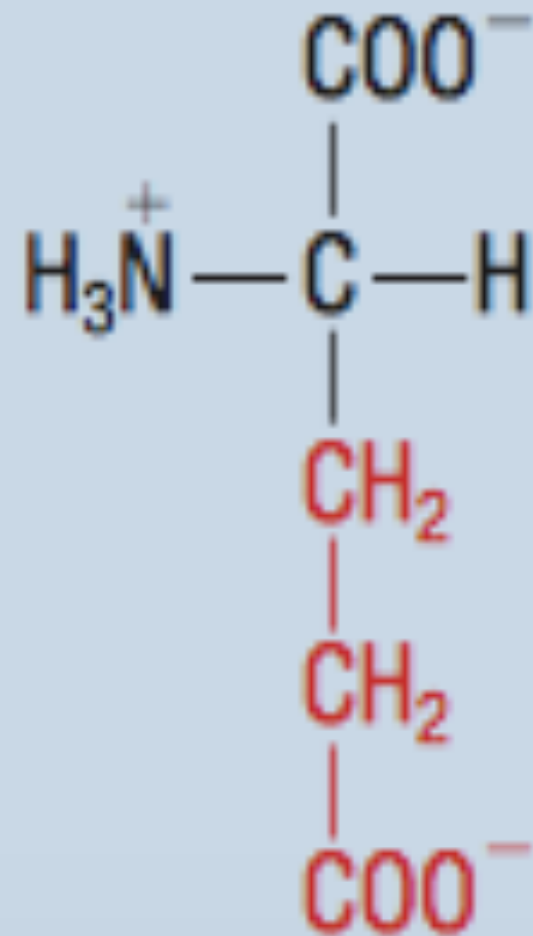


Negatively Charged R Groups

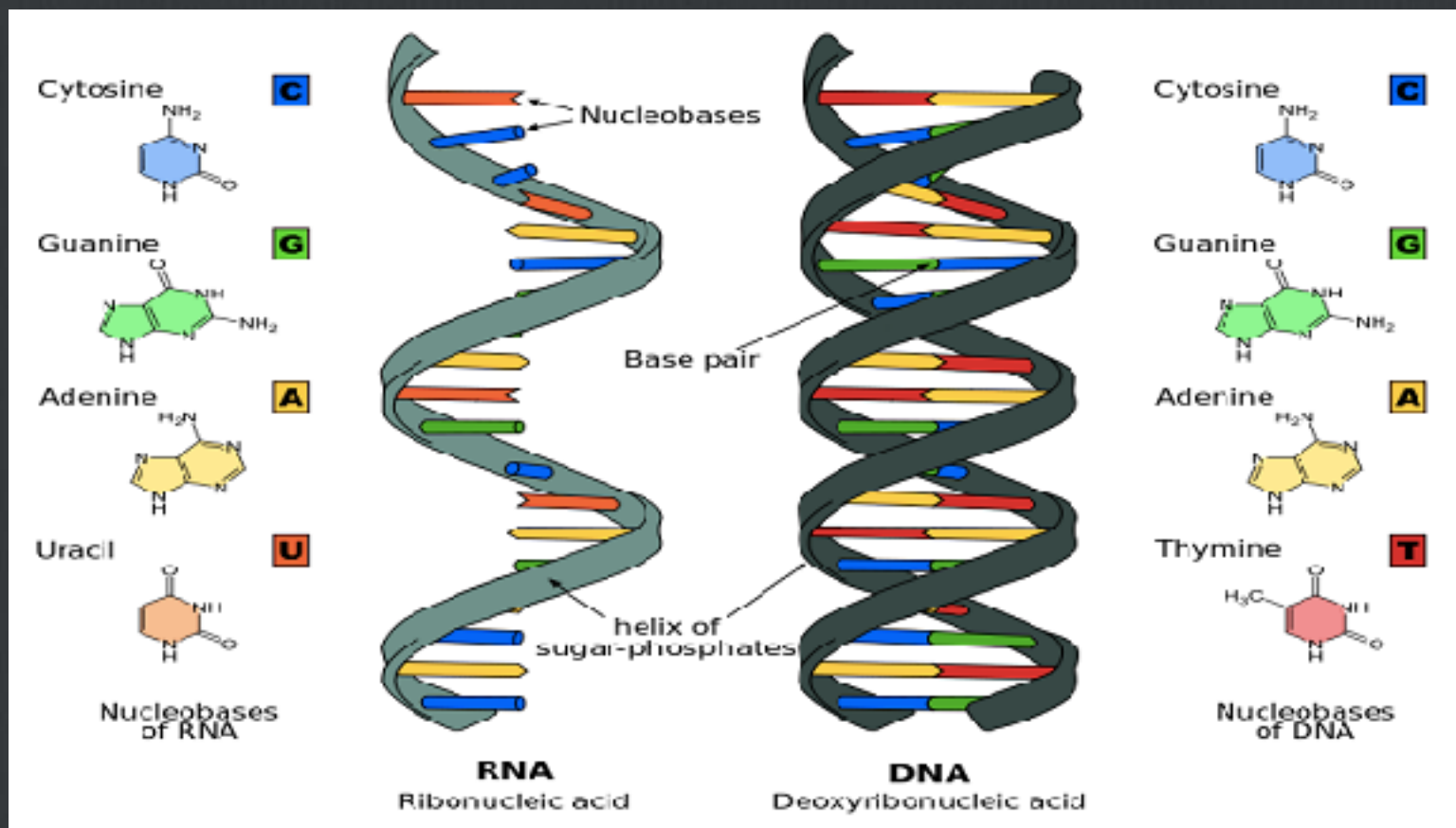
aspartate (asp)



glutamate (glu)



Nucleic Acids



DNA (deoxyribonucleic acid): a polymer in the cell nucleus that stores and transmits genetic information.

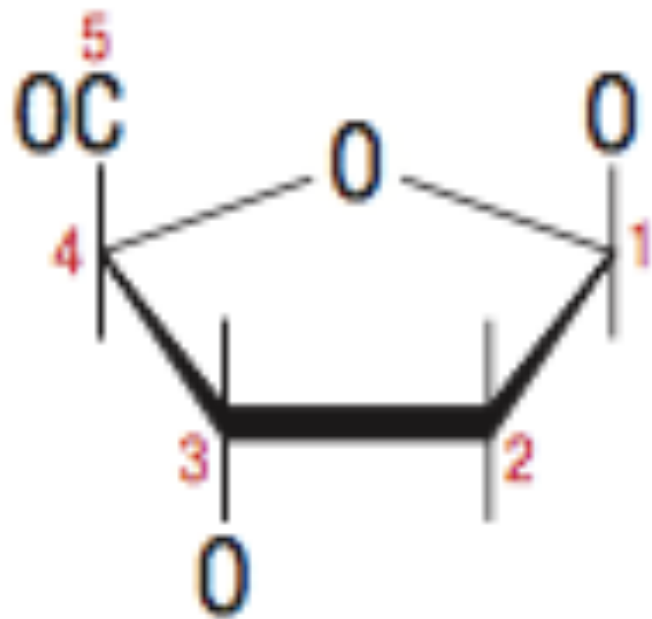
RNA (ribonucleic acid): a polymer in the cell cytoplasm that stores and transmits genetic information.

nucleic acid: the polymer molecules that make up DNA and RNA.

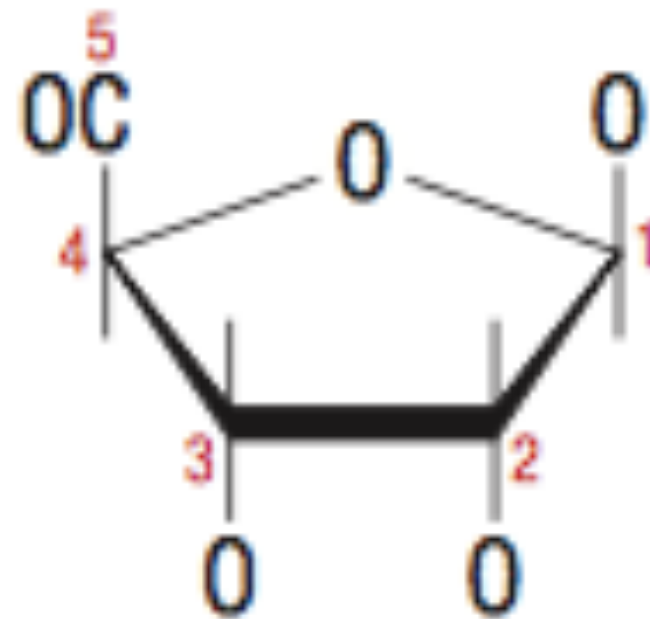
Three Parts of Nucleotides

The monomers of the nucleic acids, called **nucleotides**, are composed of three distinct parts:

1. a 5-carbon sugar: deoxyribose in DNA and ribose in RNA

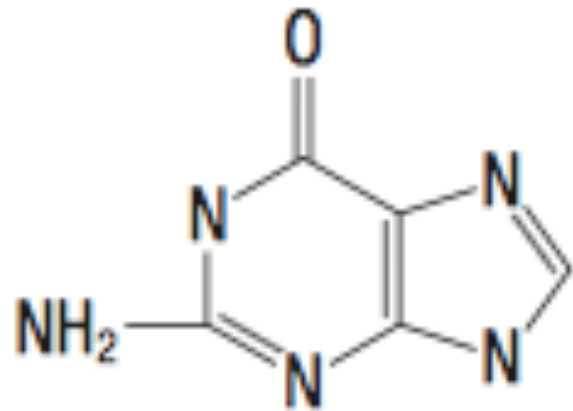


Deoxyribose

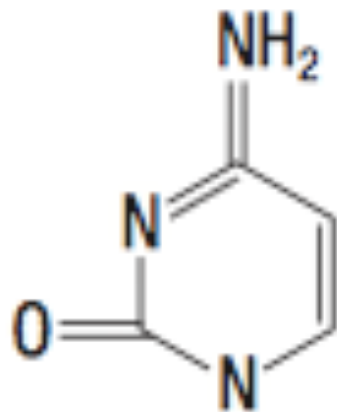


Ribose

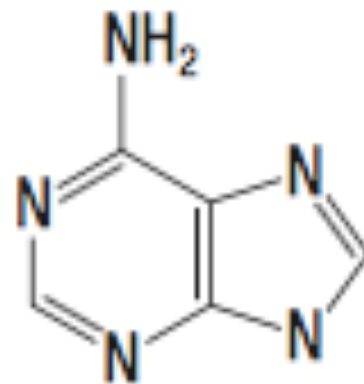
2. a **nitrogen-containing organic base**



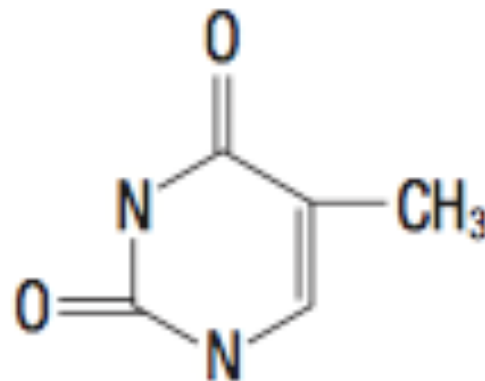
guanine (G)



cytosine (C)



adenine (A)



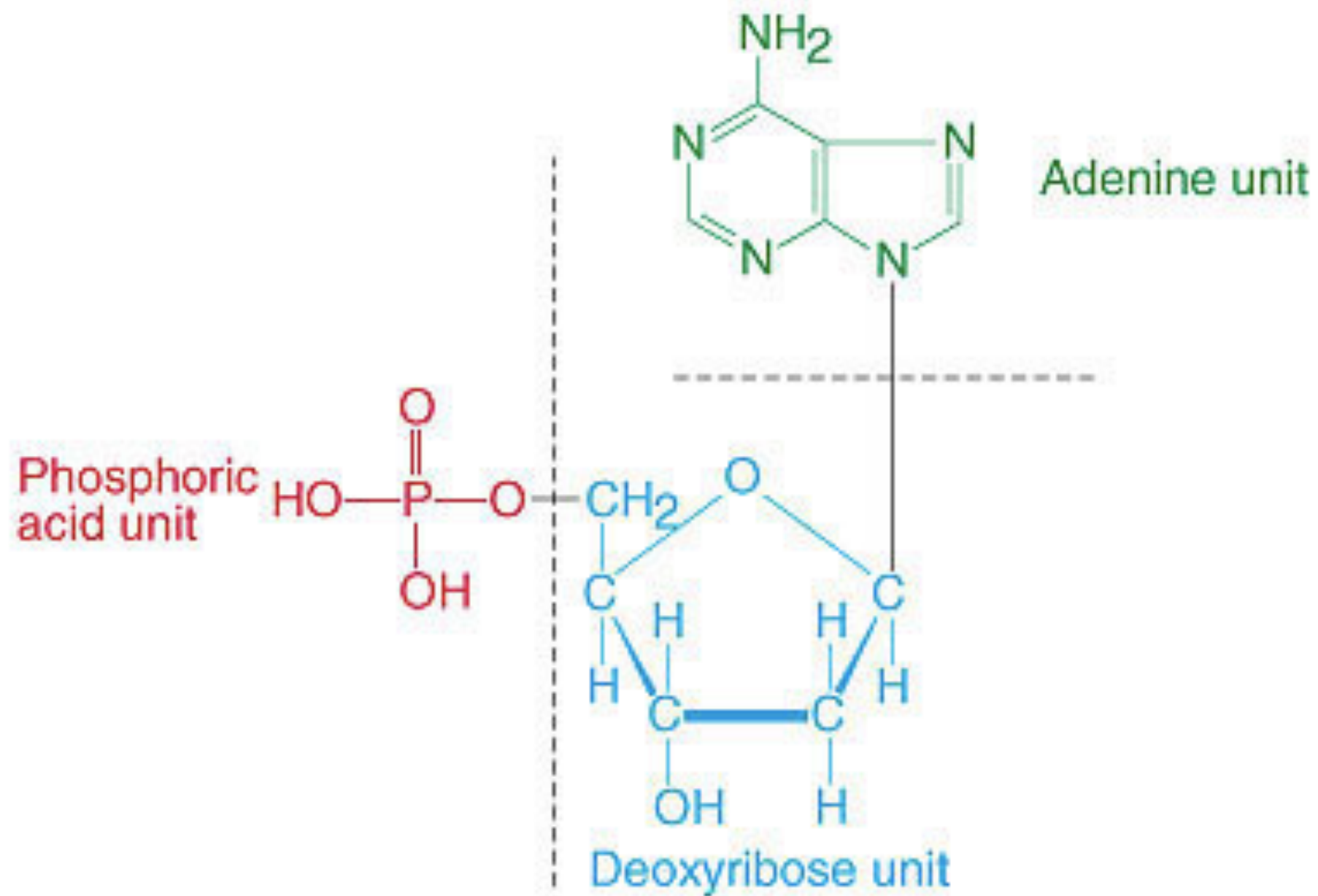
thymine (T)

The key to DNA's functioning is its **double-helical structure with complementary bases on the two strands**. The bases form hydrogen bonds with each other.

The structures of **cytosine and guanine** make them perfect partners for hydrogen bonding, and they are always found opposite each other on the two strands of DNA.

Thymine and adenine form similar hydrogen-bonding pairs.

- 3. a phosphoric acid molecule, H_3PO_4





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